

Combinatorial proof of the sum formula for q MZVs

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Today's slides

- 1 Introduction
- 2 Combinatorial interpretations
- 3 Bijection

1 Introduction

2 Combinatorial interpretations

3 Bijection

q MZVs

Definition (q MZVs (cf. Bradley, 2005[1]))

Let $\mathbf{k} = (k_1, \dots, k_r) \in (\mathbb{Z}_{>0})^r$ be an admissible index; that is, $k_r > 1$. We define (BZ-type) q MZVs as follows:

$$\zeta_q(\mathbf{k}) := \sum_{0 < m_1 < \dots < m_r} \frac{q^{m_1(k_1-1) + m_2(k_2-1) + \dots + m_r(k_r-1)}}{(1 - q^{m_1})^{k_1} (1 - q^{m_2})^{k_2} \dots (1 - q^{m_r})^{k_r}} \in \mathbb{Z}_{\geq 0}[[q]].$$

Theorem 1 (Sum formula for q MZVs (Bradley, 2005[1]))

Let k, r be integers such that $r + 1 \leq k$. Then, the following holds.

$$\sum_{\substack{k_1, \dots, k_{r-1} \in \mathbb{Z}_{\geq 1} \\ k_r \in \mathbb{Z}_{\geq 2} \\ k_1 + \dots + k_r = k}} \zeta_q(k_1, \dots, k_r) = \zeta_q(k).$$

Example and my question

$$\zeta_q(4) = 1 \times q^3 + 4 \times q^4 + 10 \times q^5 + 21 \times q^6 + \dots$$

$$\zeta_q(1, 3) = 0 \times q^3 + 1 \times q^4 + 1 \times q^5 + 6 \times q^6 + \dots$$

$$\zeta_q(2, 2) = 1 \times q^3 + 3 \times q^4 + 9 \times q^5 + 15 \times q^6 + \dots$$

My question

Can these equations be interpreted combinatorially?

- 1 Introduction
- 2 Combinatorial interpretations
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Definition of combinatorial objects

Definition

Let $\mathbf{k} = (k_1, \dots, k_r)$ be an admissible index or $\mathbf{k} = (1)$.

(1) We define the following object.

$$\mathcal{C}(\mathbf{k}) := \{(\mathbf{m}; \mathbf{u}_1; \dots; \mathbf{u}_r) \mid \text{conditions (i) and (ii) hold}\}$$

(i) $\mathbf{m} = (m_1, \dots, m_r) \in (\mathbb{Z}_{>0})^r$, with $0 < m_1 < m_2 < \dots < m_r$,

(ii) $\mathbf{u}_i = (u_{i,1}, \dots, u_{i,k_i}) \in (\mathbb{Z}_{\geq 0})^{k_i}$, with $0 \leq u_{i,1} < \dots < u_{i,k_i}$ ($i = 1, 2, \dots, r$).

(2) For $N \in \mathbb{Z}_{>0}$, we define $\mathcal{C}(\mathbf{k}; N)$ as follows:

$$\mathcal{C}(\mathbf{k}; N) := \left\{ C = (\mathbf{m}; \mathbf{u}_1; \dots; \mathbf{u}_r) \in \mathcal{C}(\mathbf{k}) \mid N(C) := \sum_{i=1}^r m_i u_{i,k_i} = N \right\}.$$

Definition of combinatorial objects

Definition

(3) For $N \in \mathbb{Z}_{>0}$, $M \in \mathbb{Z}_{\geq 0}$, we define $\mathcal{C}(\mathbf{k}; N; M)$ as follows:

$$\mathcal{C}(\mathbf{k}; N; M) := \left\{ C = (\mathbf{m}; \mathbf{u}_1; \dots; \mathbf{u}_r) \in \mathcal{C}(\mathbf{k}; N) \left| p(C) := \sum_{i=1}^r u_{i,1} = M \right. \right\}.$$

Example ($\mathcal{C}(\mathbf{k}; N)$)

$$\mathcal{C}(1, 3; 6) = \{((1, 2); (0); (0, 1, 3)), ((1, 2); (0); (0, 2, 3)), ((1, 2); (0); (1, 2, 3)), \\ ((1, 2); (2); (0, 1, 2)), ((1, 3); (0); (0, 1, 2)), ((2, 3); (0); (0, 1, 2))\}$$

$\rightsquigarrow \#\mathcal{C}(1, 3; 6) = 6$ and it matches $[q^6]\zeta_q(1, 3)$

Relation between $\zeta_q(\mathbf{k})$ and $\mathcal{C}(\mathbf{k})$

Proposition 2

Let $\mathbf{k}$ be an admissible index. Then,

$$\zeta_q(\mathbf{k}) = \sum_{N>0} \#\mathcal{C}(\mathbf{k}; N) q^N$$

holds.

Relation between $\zeta_q(\mathbf{k})$ and $\mathcal{C}(\mathbf{k})$

Sketch of proof.

$$\zeta_q(\mathbf{k}) = \sum_{0 < m_1 < \dots < m_r} \prod_{i=1}^r \frac{q^{m_i(k_i-1)}}{(1-q^{m_i})^{k_i}} = \sum_{0 < m_1 < \dots < m_r} \sum_{0 \leq n_1, \dots, n_r} \prod_{i=1}^r \binom{n_i}{k_i-1} q^{n_i m_i}$$

$$\begin{aligned} \rightsquigarrow [q^N] \zeta_q(\mathbf{k}) &= \sum_{\substack{0 < m_1 < \dots < m_r \\ 0 \leq n_1, \dots, n_r \\ m_1 n_1 + \dots + m_r n_r = N}} \prod_{i=1}^r \binom{n_i}{k_i-1} \\ &= \sum_{i=1}^r \prod_{i=1}^r \#\{(u_{i,1}, \dots, u_{i,k_i}) \in (\mathbb{Z}_{\geq 0})^{k_i} \mid 0 \leq u_{i,1} < \dots < u_{i,k_i} = n_i\} \\ &= \#\mathcal{C}(\mathbf{k}; N) \end{aligned}$$

Roles of $p(C)$

Definition

For $C = (\mathbf{m}; \mathbf{u}_1; \dots; \mathbf{u}_r) \in \mathcal{C}(\mathbf{k})$ and $L \in \mathbb{Z}_{\geq 0}$, we define $\mathcal{E}(C, L)$ as follows:

$$\mathcal{E}(C, L) = \{(\mathbf{m}; \mathbf{v}_1; \dots; \mathbf{v}_r) \mid \text{condition } (*) \text{ holds}\}$$

(*) $\mathbf{v}_i = (v_{i,1}, \dots, v_{i,|\mathbf{v}_i|}) \in (\mathbb{Z}_{\geq 0})^{|\mathbf{v}_i|}$, $l_i \in \mathbb{Z}_{\geq 1}$ satisfying the following conditions:

- ▶ $k_i \leq |\mathbf{v}_i|$ ($i = 1, 2, \dots, r$),
- ▶ $\sum_i (|\mathbf{v}_i| - k_i) = L$,
- ▶ $0 \leq v_{i,1} < \dots < v_{i,|\mathbf{v}_i|}$ ($i = 1, 2, \dots, r$),
- ▶ $v_{i,j+|\mathbf{v}_i|-k_i} = u_{i,j}$ ($i = 1, 2, \dots, r$, $j = 1, 2, \dots, k_i$). (i.e. u_i is suffix of v_i)

Roles of $p(C)$

Lemma 3

1. $\#\mathcal{E}(C, L)$ only depends on $p(C)$ and L .

It is independent of the specific components of C and $\mathbf{k}$.

2. Let $(k_1, \dots, k_r)$ be an admissible index and $N \in \mathbb{Z}_{>0}$, $L \in \mathbb{Z}_{\geq 0}$. Then,

$$\bigsqcup_{C \in \mathcal{C}(k_1, \dots, k_r; N)} \mathcal{E}(C, L) = \bigsqcup_{\substack{l_1 + \dots + l_r = L \\ l_i \geq 0}} \mathcal{C}(k_1 + l_1, \dots, k_r + l_r; N)$$

holds.

Roles of $p(C)$

Denote $\#\mathcal{E}(C, L)$ by $f(p(C), L)$.

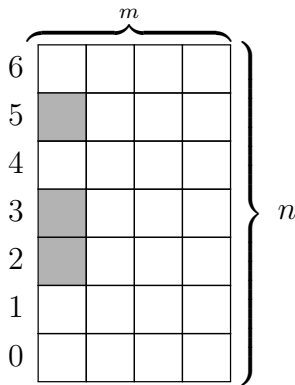
$$\begin{aligned}\sum_{\substack{l_1+\dots+l_r=L \\ l_i \geq 0}} \zeta_q(k_1 + l_1, \dots, k_r + l_r) &= \sum_{N>0} \sum_{\substack{l_1+\dots+l_r=L \\ l_i \geq 0}} \#\mathcal{C}(k_1 + l_1, \dots, k_r + l_r; N) q^N \\ &= \sum_{N>0} \sum_{C \in \mathcal{C}(k_1, \dots, k_r; N)} \#\mathcal{E}(C, L) q^N \\ &= \sum_{\substack{N>0 \\ M \geq 0}} \sum_{C \in \mathcal{C}(k_1, \dots, k_r; N; M)} \#\mathcal{E}(C, L) q^N \\ &= \sum_{\substack{N>0 \\ M \geq 0}} \#\mathcal{C}(k_1, \dots, k_r; N; M) \times f(M, L) q^N.\end{aligned}$$

Then, once we establish $\#\mathcal{C}((\{1\}^{k-1}, 2); N; M) = \#\mathcal{C}((k+1); N; M)$ for all $N \in \mathbb{Z}_{>0}$ and $M \in \mathbb{Z}_{\geq 0}$, the sum formula follows.

Marked partition $\mathcal{C}$ (c.f. Brindle[2])

Let $k \geq 1$, $C = ((m); (u_{1,1}, u_{1,2}, \dots, u_{1,k})) \in \mathcal{C}(k)$.

Visualize C as follows:



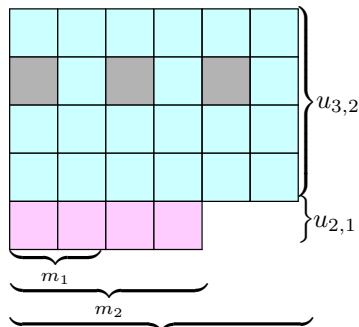
- ▶ A grid with $u_{1,k}$ rows and m columns.
- ▶ Place marks (filled with gray) on the $u_1, u_2, \dots, u_{k-1}$ -th cells from the bottom in the leftmost column (0-indexed).

$$((m); (u_{1,1}, u_{1,2}, u_{1,3}, u_{1,4})) = ((4); (2, 3, 5, 7)) \in \mathcal{C}(4; 28)$$

Marked partition $\mathcal{C}(\{1\}^{r-1}, 2)$ (c.f. Brindle[2])

Let $r \geq 1$, $C = ((m_1, m_2, \dots, m_r); (u_{1,1}); (u_{2,1}); \dots; (u_{r,1}, u_{r,2})) \in \mathcal{C}(\{1\}^{r-1}, 2)$.

Visualize C as follows:



- ▶ Stack $u_{r,2} \times m_r$, $u_{r-1,1} \times m_{r-1}$, $\dots$, $u_{1,1} \times m_1$ grids vertically.
- ▶ Place marks (filled with gray) on the $0, m_1, \dots, m_{r-1}$ -th cells from the left in the $u_{r,2} - u_{r,1} - 1$ -th rows from the top (0-indexed).

$$((m_1, m_2, m_3); (u_{1,1}); (u_{2,1}); (u_{3,1}, u_{3,2})) = ((2, 4, 6); (0), (1), (2, 4)) \in \mathcal{C}((1, 1, 2); 28)$$

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Today's goal

Constructing an explicit bijection between $\mathcal{C}(k+1)$ and $\mathcal{C}(\{1\}^{k-1}, 2)$

- ▶ If the bijection preserves $N(C) \rightsquigarrow \zeta_q(k+1) = \zeta_q(\{1\}^{k-1}, 2)$ ($\because$ Proposition 2)
- ▶ If the bijection preserves both $N(C)$ and $p(C) \rightsquigarrow$ sum formula. ($\because$ Lemma 3)

Before explaining the details...

Examples of correspondences



Reformulating $\mathcal{C}(k)$

For convenience, we reformulate $\mathcal{C}(k)$ as follows:

Definition

$\mathcal{C}(k) := \{(H, W, U) \mid H, W \in \mathbb{Z}_{>0}, U = (u_1, \dots, u_{k-1}) \in (\mathbb{Z}_{\geq 0})^{k-1}, 0 \leq u_1 < \dots < u_{k-1} < H\}.$

Remark 4

The above definition is essentially the same as the $\text{dep}(k) = 1$ case of the previous definition.

$$((m_1); (u_{1,1}, \dots, u_{1,k-1}, u_{1,k})) \leftrightarrow (u_{1,k}, m_1, (u_{1,1}, \dots, u_{1,k-1}))$$

Reformulating $\mathcal{C}(k)$

Definition

For each $C = (H, W, U) \in \mathcal{C}(k)$, we write $H(C) = H$, $W(C) = W$, $U(C) = U$.

Remark 5

$$\mathcal{C}(3) \ni (3, 4, (0, 2)) \longleftrightarrow$$

2			
1			
0			

- ▶ $H(C)$ is the “height” of C .
- ▶ $W(C)$ is the “width” of C .
- ▶ $U(C)$ is the “position of marks” of C .

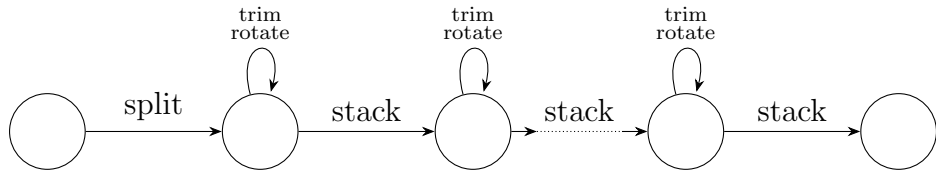
Constructing a bijection from $\mathcal{C}(\{1\}^{k-1}, 2)$ to $\mathcal{C}(k+1)$

overview

We use four operations “Splitting”, “Trimming”, “Rotating” and “Stacking”.

(STEP 1) Apply “Splitting” to $\mathcal{C}(\{1\}^{k-1}, 2)$.

(STEP 2) Repeat “Trimming”, “Rotating” and “Stacking”.



Splitting operation

Definition 6 (Splitting operation)

We define a map $\text{split} : \mathcal{C}(\{1\}^{k-1}, 2) \rightarrow \mathcal{C}(2) \times \mathcal{C}(1) \times \cdots \times \mathcal{C}(1)$ as follows:

$$\begin{aligned} \mathcal{C}(\{1\}^{k-1}, 2) \ni ((m_1, m_2, \dots, m_k); (u_{1,1}); \dots; (u_{k-1,1}); (u_{k,1}, u_{k,2})) \\ \xrightarrow{\text{split}} (C_1, C_2, \dots, C_k) \in \mathcal{C}(2) \times \mathcal{C}(1) \times \cdots \times \mathcal{C}(1) \end{aligned}$$

where,

$$C_1 = (m_1, u_{1,1} + \cdots + u_{k,2}, (u_{1,1} + \cdots + u_{k-1,1} + u_{k,1})) \in \mathcal{C}(2)$$

$$C_2 = (m_2 - m_1, u_{2,1} \cdots + u_{k,2}, \emptyset) \in \mathcal{C}(1)$$

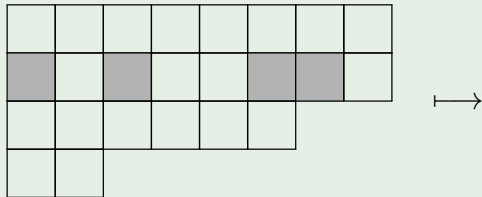
$$\vdots$$

$$C_k = (m_k - m_{k-1}, u_{k,2}, \emptyset) \in \mathcal{C}(1)$$

Visualizing the splitting operation

Example of splitting

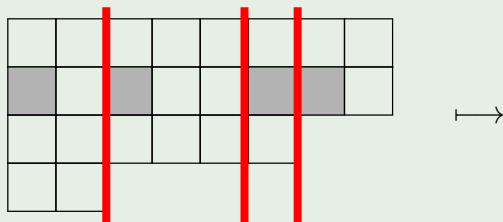
$$((2, 5, 6, 8); (1); (0); (1); (0, 2)) \xrightarrow{\text{split}} ((4, 2, (2)), (3, 3, \emptyset), (3, 1, \emptyset), (2, 2, \emptyset))$$



Visualizing the splitting operation

Example of splitting

$$((2, 5, 6, 8); (1); (0); (1); (0, 2)) \xrightarrow{\text{split}} ((4, 2, (2)), (3, 3, \emptyset), (3, 1, \emptyset), (2, 2, \emptyset))$$



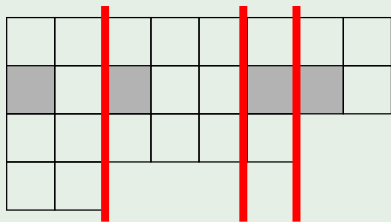
Visualizing the splitting operation

Example of splitting

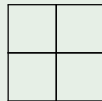
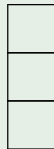
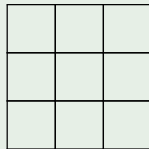
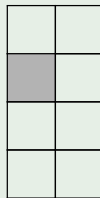
$((2, 5, 6, 8); (1); (0); (1); (0, 2))$

$\xrightarrow{\text{split}}$

$((4, 2, (2)), (3, 3, \emptyset), (3, 1, \emptyset), (2, 2, \emptyset))$



$\mapsto$



Same and Different Widths

Definition 7

For $k, l \in \mathbb{Z}_{>0}$, we partition $\mathcal{C}(k) \times \mathcal{C}(l)$ into subsets based on whether they have the “same width” or “different widths” as follows:

$$\mathcal{C}_{\text{sw}}(k, l) := \{((H_1, W_1, U_1), (H_2, W_2, U_2)) \in \mathcal{C}(k) \times \mathcal{C}(l) \mid W_1 = W_2\},$$

$$\mathcal{C}_{\text{dw}}(k, l) := \{((H_1, W_1, U_1), (H_2, W_2, U_2)) \in \mathcal{C}(k) \times \mathcal{C}(l) \mid W_1 \neq W_2\}.$$

Trimming operation

Definition 8 (Trimming operation)

We define a map $\text{trim} : \mathcal{C}_{\text{dw}}(k, 1) \rightarrow \mathcal{C}(k) \times \mathcal{C}(1)$ as follows:

$$\begin{array}{ccc} \text{trim} : \mathcal{C}_{\text{dw}}(k, 1) & \rightarrow & \mathcal{C}(k) \times \mathcal{C}(1) \\ \Psi & & \Psi \\ ((H_1, W_1, U_1), (H_2, W_2, \emptyset)) & \mapsto & \begin{cases} ((H_1, W_1 - W_2, U_1), (H_2 + H_1, W_2, \emptyset)) & (W_1 > W_2) \\ ((H_1 + H_2, W_1, U_1), (H_1, W_2 - W_1, \emptyset)) & (W_1 < W_2) \end{cases} \end{array}$$

Visualizing the trimming operation

$$((H_1, W_1, U_1), (H_2, W_2, \emptyset)) \xrightarrow{\text{trim}} \begin{cases} ((H_1, W_1 - W_2, U_1), (H_2 + H_1, W_2, \emptyset)) & (W_1 > W_2) \\ ((H_1 + H_2, W_1, U_1), (H_1, W_2 - W_1, \emptyset)) & (W_1 < W_2) \end{cases}$$

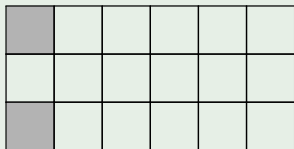
Example of trimming for

Visualizing the trimming operation

$$((H_1, W_1, U_1), (H_2, W_2, \emptyset)) \xrightarrow{\text{trim}} \begin{cases} ((H_1, W_1 - W_2, U_1), (H_2 + H_1, W_2, \emptyset)) & (W_1 > W_2) \\ ((H_1 + H_2, W_1, U_1), (H_1, W_2 - W_1, \emptyset)) & (W_1 < W_2) \end{cases}$$

Example of trimming for $(W_1 > W_2)$

$$\mathcal{C}_{\text{dw}}(3, 1) \ni ((3, 6, (0, 2)), (1, 2)) \xrightarrow{\text{trim}} ((3, 4, (0, 2)), (4, 2)) \in \mathcal{C}(3) \times \mathcal{C}(1)$$

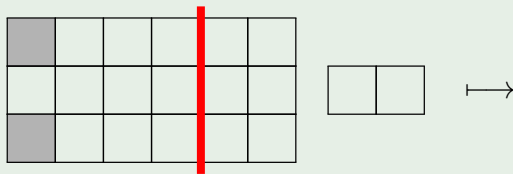


Visualizing the trimming operation

$$((H_1, W_1, U_1), (H_2, W_2, \emptyset)) \xrightarrow{\text{trim}} \begin{cases} ((H_1, W_1 - W_2, U_1), (H_2 + H_1, W_2, \emptyset)) & (W_1 > W_2) \\ ((H_1 + H_2, W_1, U_1), (H_1, W_2 - W_1, \emptyset)) & (W_1 < W_2) \end{cases}$$

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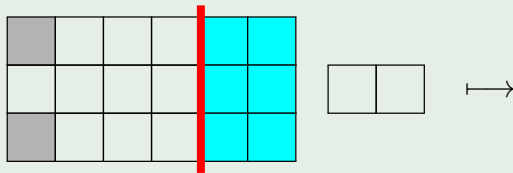


Visualizing the trimming operation

$$((H_1, W_1, U_1), (H_2, W_2, \emptyset)) \xrightarrow{\text{trim}} \begin{cases} ((H_1, W_1 - W_2, U_1), (H_2 + H_1, W_2, \emptyset)) & (W_1 > W_2) \\ ((H_1 + H_2, W_1, U_1), (H_1, W_2 - W_1, \emptyset)) & (W_1 < W_2) \end{cases}$$

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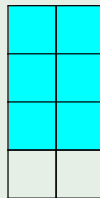
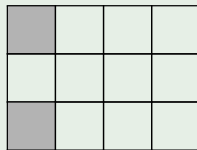
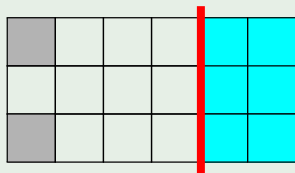


Visualizing the trimming operation

$$((H_1, W_1, U_1), (H_2, W_2, \emptyset)) \xrightarrow{\text{trim}} \begin{cases} ((H_1, W_1 - W_2, U_1), (H_2 + H_1, W_2, \emptyset)) & (W_1 > W_2) \\ ((H_1 + H_2, W_1, U_1), (H_1, W_2 - W_1, \emptyset)) & (W_1 < W_2) \end{cases}$$

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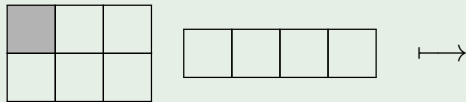


Visualizing the trimming operation

$$((H_1, W_1, U_1), (H_2, W_2, \emptyset)) \xrightarrow{\text{trim}} \begin{cases} ((H_1, W_1 - W_2, U_1), (H_2 + H_1, W_2, \emptyset)) & (W_1 > W_2) \\ ((H_1 + H_2, W_1, U_1), (H_1, W_2 - W_1, \emptyset)) & (W_1 < W_2) \end{cases}$$

Example of trimming for $(W_1 < W_2)$

$$\mathcal{C}_{\text{dw}}(2, 1) \ni ((2, 3, (1)), (1, 4)) \xrightarrow{\text{trim}} ((3, 3, (1)), (1, 1)) \in \mathcal{C}(2) \times \mathcal{C}(1)$$

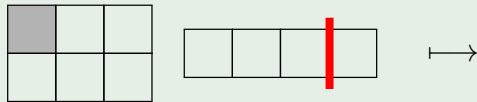


Visualizing the trimming operation

$$((H_1, W_1, U_1), (H_2, W_2, \emptyset)) \xrightarrow{\text{trim}} \begin{cases} ((H_1, W_1 - W_2, U_1), (H_2 + H_1, W_2, \emptyset)) & (W_1 > W_2) \\ ((H_1 + H_2, W_1, U_1), (H_1, W_2 - W_1, \emptyset)) & (W_1 < W_2) \end{cases}$$

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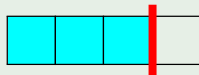
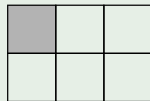


Visualizing the trimming operation

$$((H_1, W_1, U_1), (H_2, W_2, \emptyset)) \xrightarrow{\text{trim}} \begin{cases} ((H_1, W_1 - W_2, U_1), (H_2 + H_1, W_2, \emptyset)) & (W_1 > W_2) \\ ((H_1 + H_2, W_1, U_1), (H_1, W_2 - W_1, \emptyset)) & (W_1 < W_2) \end{cases}$$

Example of trimming for $(W_1 < W_2)$

$$\mathcal{C}_{\text{dw}}(2, 1) \ni ((2, 3, (1)), (1, 4)) \xrightarrow{\text{trim}} ((3, 3, (1)), (1, 1)) \in \mathcal{C}(2) \times \mathcal{C}(1)$$

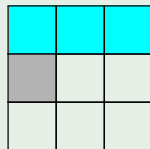
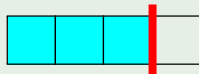
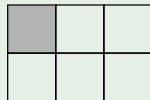


Visualizing the trimming operation

$$((H_1, W_1, U_1), (H_2, W_2, \emptyset)) \xrightarrow{\text{trim}} \begin{cases} ((H_1, W_1 - W_2, U_1), (H_2 + H_1, W_2, \emptyset)) & (W_1 > W_2) \\ ((H_1 + H_2, W_1, U_1), (H_1, W_2 - W_1, \emptyset)) & (W_1 < W_2) \end{cases}$$

Example of trimming for $(W_1 < W_2)$

$$\mathcal{C}_{\text{dw}}(2, 1) \ni ((2, 3, (1)), (1, 4)) \xrightarrow{\text{trim}} ((3, 3, (1)), (1, 1)) \in \mathcal{C}(2) \times \mathcal{C}(1)$$



Stacking operation

Definition 9 (Stacking operation)

We define a map $\text{stack} : \mathcal{C}_{\text{sw}}(k, 1) \rightarrow \mathcal{C}(k + 1)$ as follows:

$$\begin{array}{ccc} \text{stack} : \mathcal{C}_{\text{sw}}(k, 1) & \rightarrow & \mathcal{C}(k + 1) \\ \Downarrow & & \Downarrow \\ ((H_1, W, (u_1, \dots, u_{k-1})), (H_2, W, \emptyset)) & \mapsto & (H_1 + H_2, W, (u_1, \dots, u_{k-1}, H_1)) \end{array}$$

Visualizing the stacking operation

$$((H_1, W, (u_1, \dots, u_{k-1})), (H_2, W, \emptyset)) \xrightarrow{\text{stack}} (H_1 + H_2, W, (u_1, \dots, u_{k-1}, H_1))$$

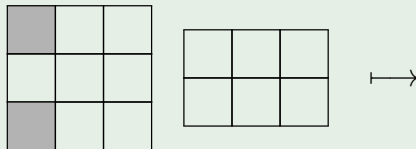
Example of stacking

Visualizing the stacking operation

$$((H_1, W, (u_1, \dots, u_{k-1})), (H_2, W, \emptyset)) \xrightarrow{\text{stack}} (H_1 + H_2, W, (u_1, \dots, u_{k-1}, H_1))$$

Example of stacking

$$\mathcal{C}_{\text{sw}}(3, 1) \ni ((3, 3, (0, 2)), (2, 3)) \xrightarrow{\text{stack}} (5, 3, (0, 2, 3)) \in \mathcal{C}(4)$$

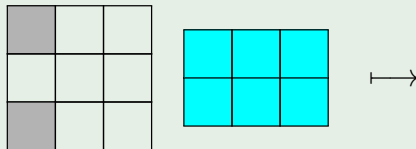


Visualizing the stacking operation

$$((H_1, W, (u_1, \dots, u_{k-1})), (H_2, W, \emptyset)) \xrightarrow{\text{stack}} (H_1 + H_2, W, (u_1, \dots, u_{k-1}, H_1))$$

Example of stacking

$$\mathcal{C}_{\text{sw}}(3, 1) \ni ((3, 3, (0, 2)), (2, 3)) \xrightarrow{\text{stack}} (5, 3, (0, 2, 3)) \in \mathcal{C}(4)$$

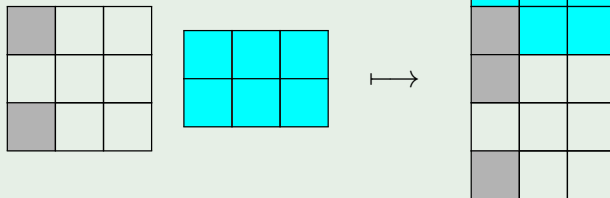


Visualizing the stacking operation

$$((H_1, W, (u_1, \dots, u_{k-1})), (H_2, W, \emptyset)) \xrightarrow{\text{stack}} (H_1 + H_2, W, (u_1, \dots, u_{k-1}, H_1))$$

Example of stacking

$$\mathcal{C}_{\text{sw}}(3, 1) \ni ((3, 3, (0, 2)), (2, 3)) \xrightarrow{\text{stack}} (5, 3, (0, 2, 3)) \in \mathcal{C}(4)$$



Rotating operation

Definition 10 (Rotating operation)

We define a map $\text{rotate} : \mathcal{C}(2) \times \mathcal{C}(1) \times \cdots \times \mathcal{C}(1) \rightarrow \mathcal{C}(2) \times \mathcal{C}(1) \times \cdots \times \mathcal{C}(1)$ as follows:

$$\begin{array}{ccc} \text{rotate} : \mathcal{C}(2) \times \mathcal{C}(1) \times \cdots \times \mathcal{C}(1) & \rightarrow & \mathcal{C}(2) \times \mathcal{C}(1) \times \cdots \times \mathcal{C}(1) \\ \Psi & & \Psi \\ (C_1, C_2, C_3, \dots, C_{k-1}, C_k) & \mapsto & (C_1, C_k, C_2, C_3, \dots, C_{k-1}) \end{array}$$

✂ We don't move C_1 .

Visualizing the rotating operation

$$(C_1, C_2, C_3, \dots, C_{k-1}, C_k) \xrightarrow{\text{rotate}} (C_1, C_k, C_2, C_3, \dots, C_{k-1})$$

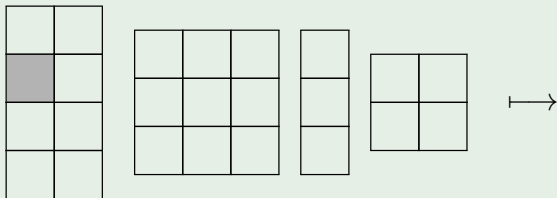
Example of rotating

Visualizing the rotating operation

$$(C_1, C_2, C_3, \dots, C_{k-1}, C_k) \xrightarrow{\text{rotate}} (C_1, C_k, C_2, C_3, \dots, C_{k-1})$$

Example of rotating

$$((4, 2, (2)), (3, 3, \emptyset), (3, 1, \emptyset), (2, 2, \emptyset)) \xrightarrow{\text{rotate}} ((4, 2, (2)), (2, 2, \emptyset), (3, 3, \emptyset), (3, 1, \emptyset))$$

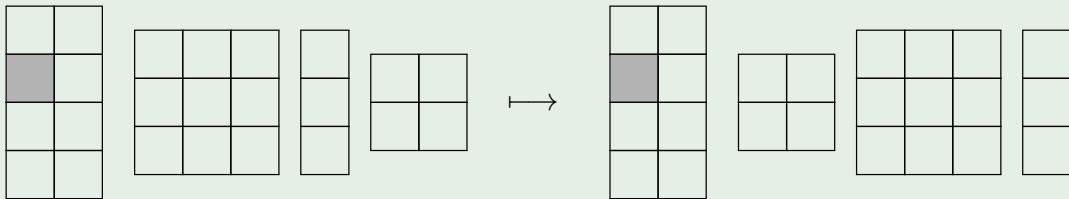


Visualizing the rotating operation

$$(C_1, C_2, C_3, \dots, C_{k-1}, C_k) \xrightarrow{\text{rotate}} (C_1, C_k, C_2, C_3, \dots, C_{k-1})$$

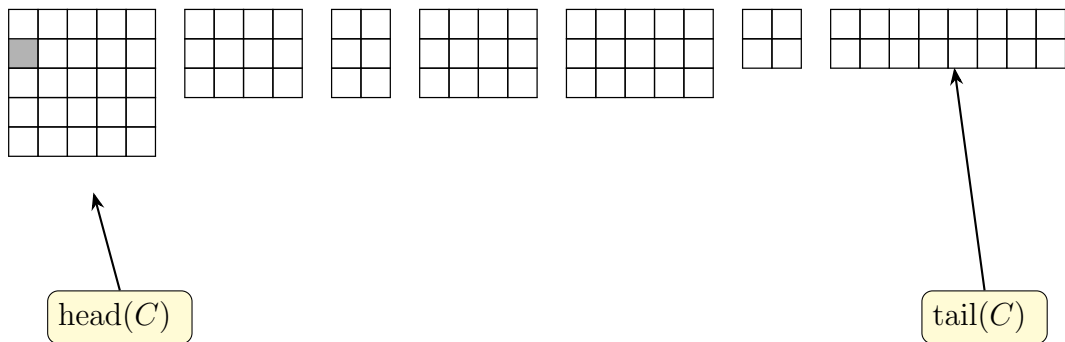
Example of rotating

$$((4, 2, (2)), (3, 3, \emptyset), (3, 1, \emptyset), (2, 2, \emptyset)) \xrightarrow{\text{rotate}} ((4, 2, (2)), (2, 2, \emptyset), (3, 3, \emptyset), (3, 1, \emptyset))$$



Head and Tail

For $C = (C_1, \dots, C_t) \in \mathcal{C}(2) \times \mathcal{C}(1) \times \dots \times \mathcal{C}(1)$, we call C_1 the head (denote $\text{head}(C)$) and C_t the tail (denote $\text{tail}(C)$).

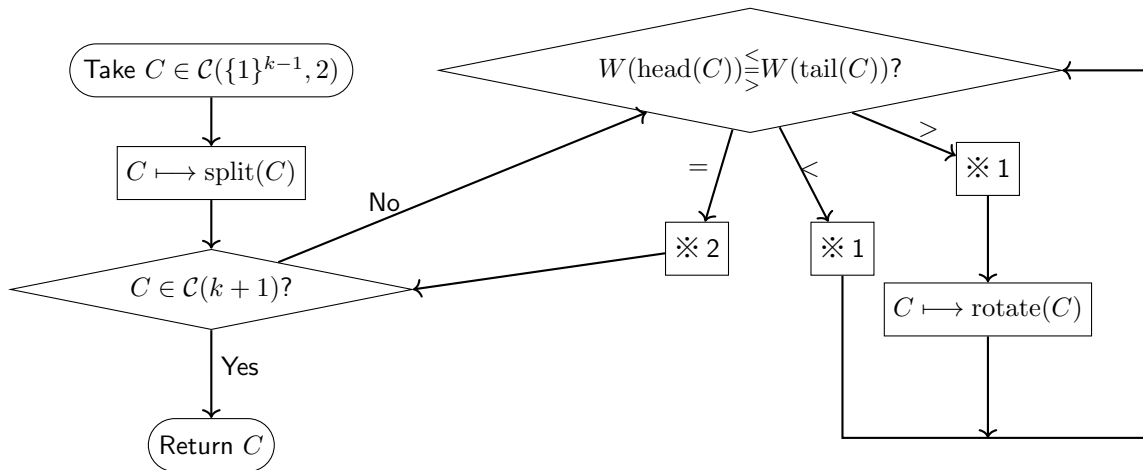


Map from $\mathcal{C}(\{1\}^{k-1}, 2)$ to $\mathcal{C}(k+1)$

For each $C \in \mathcal{C}(\{1\}^{k-1}, 2)$, we apply the following procedure:

1. Replace C with $\text{split}(C)$, and go to 2.
2. If $C \in \mathcal{C}(k+1)$, return C and end the procedure; otherwise, go to 3.
3. Compare $W(\text{head}(C))$ and $W(\text{tail}(C))$:
 - 3.1 If $W(\text{head}(C)) < W(\text{tail}(C))$, replace $\text{head}(C)$ and $\text{tail}(C)$ with $\text{trim}(\text{head}(C), \text{tail}(C))$. Then, go to 3.
 - 3.2 If $W(\text{head}(C)) > W(\text{tail}(C))$, replace $\text{head}(C)$ and $\text{tail}(C)$ with $\text{trim}(\text{head}(C), \text{tail}(C))$. Then, replace C with $\text{rotate}(C)$ and go to 3.
 - 3.3 If $W(\text{head}(C)) = W(\text{tail}(C))$, replace $\text{head}(C)$ and $\text{tail}(C)$ with $\text{stack}(\text{head}(C), \text{tail}(C))$. Then, go to 2.

Flow chart



※ 1: " $(\text{head}(C), \text{tail}(C)) \mapsto \text{trim}(\text{head}(C), \text{tail}(C))$ "

※ 2: " $(\text{head}(C), \text{tail}(C)) \mapsto \text{stack}(\text{head}(C), \text{tail}(C))$ "

Example

Example of correspondences



Things to Check

Today's goal

Constructing an explicit bijection between $\mathcal{C}(k+1)$ and $\mathcal{C}(\{1\}^{k-1}, 2)$

- ▶ If the bijection preserves $N(C) \rightsquigarrow \zeta_q(k+1) = \zeta_q(\{1\}^{k-1}, 2)$
- ▶ If the bijection preserves both $N(C)$ and $p(C) \rightsquigarrow$ sum formula.

What we want to say.

- ▶ The above procedure will terminate.
- ▶ The above procedure is bijection.
- ▶ The above procedure preserves both $N(C)$ and $p(C)$.

Proof of Termination

Proposition 11

The procedure described above will terminate.

Sketch of proof.

- ▶ The operation `split` appears exactly once.

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Sketch of proof.

- ▶ The operation `split` appears exactly once.
- ▶ The operations `stack` and `trim` both strictly decrease the sum of W .

Proof of Termination

Proposition 11

The procedure described above will terminate.

Sketch of proof.

- ▶ The operation `split` appears exactly once.
- ▶ The operations `stack` and `trim` both strictly decrease the sum of W .
- ▶ Every occurrence of the operation `rotate` is always preceded by `trim`.



Injectivity

To prove injectivity, it is sufficient to show the following theorem, since it is easy to recover the state before an operation from the operation performed and the resulting state.

Theorem 12

Let $\mathcal{C}(\{1\}^{k-1}, 2) \ni S_0 \mapsto S_1 \mapsto \dots \mapsto S_t \in \mathcal{C}(k+1)$ be the sequence of results of the operations (i.e., S_{i+1} is the result of applying some operation to S_i).

Based solely on $S_i, S_{i+1}, \dots, S_t$, we can identify what type of operation was performed to transition from S_{i-1} to S_i for all $i = 1, 2, \dots, t$.

Sketch of proof of injectivity

Let $\mathcal{C}(\{1\}^{k-1}, 2) \ni S_0 \mapsto S_1 \mapsto \cdots \mapsto S_t \in \mathcal{C}(k+1)$ be the sequence of results of the operations. Let $S_i = (C_1^{(i)}, C_2^{(i)}, \dots, C_{|S_i|}^{(i)})$.

Proposition 13

(a) If S_i satisfies

- ▶ $|S_i| = k$ and,
- ▶ $H(C_1^{(i)}) \geq H(C_2^{(i)}) \geq \cdots \geq H(C_k^{(i)})$ and,
- ▶ $H(C_k^{(i)}) \geq H(C_1^{(i)}) - U(C_1^{(i)})_1$.

Then, S_i is the state after the split operation.

Sketch of proof of injectivity

Proposition 13 (continued)

(b) If S_i satisfies

▶ $|S_i| = 1$.

Then, S_i is the state after the stack operation.

(c) If S_i satisfies

▶ $H(C_1^{(i)}) < H(C_2^{(i)})$.

Then, S_i is the state after the rotate operation and S_{i-1} is the state after the trim operation.

Sketch of proof of injectivity

Proposition 13 (continued)

(d) If S_i satisfies

- ▶ $1 < |S_i| < k$,
- ▶ $H(C_1^{(i)}) \geq H(C_2^{(i)})$ and,
- ▶ $H(C_1^{(i)}) - U(C_1^{(i)})_{|U(C_1^{(i)})|} \leq H(C_{|S_i|-1}^{(i)})$.

Then, S_i is the state after the stack operation.

(e) If S_i satisfies

- ▶ $1 < |S_i| < k$,
- ▶ $H(C_1^{(i)}) \geq H(C_2^{(i)})$ and,
- ▶ $H(C_1^{(i)}) - U(C_1^{(i)})_{|U(C_1^{(i)})|} > H(C_{|S_i|-1}^{(i)})$.

Then, S_i is the state after the trim operation.

Sketch of proof of injectivity

Lemma 14

If S_i is not the state before rotate, the following hold.

$$H(C_{|S_i|}^{(i)}) + H(C_1^{(i)}) \geq H(C_2^{(i)}) \geq H(C_3^{(i)}) \geq \dots \geq H(C_{|S_i|}^{(i)})$$

Sketch of proof of Lemma 14.

By induction.

(base case): The operation from $S_0 \rightarrow S_1$ is split and it is easy to check S_1 satisfies the inequality.

(induction step): Apply each operation to S_i such that inequality holds. □

Surjectivity · Preservation of $N(C)$, $p(C)$

Surjectivity can be verified as follows:

Theorem 15

Let $S \in \mathcal{C}(k+1)$. Then, by applying the criteria in Proposition 13, we can trace the sequence of operations back to $\mathcal{C}(\{1\}^{k-1}, 2)$.

Preservation can also be verified as follows:

Theorem 16

The procedure preserves both $N(C)$ and $p(C)$.

$\therefore$ “The total number of cells” and “the number of cells lying below the lowest mark in the head” are preserved by every operation.

Conclusion

To summarize, the procedure we have constructed satisfies the following properties:

- ▶ It is a bijection from $\mathcal{C}(\{1\}^{k-1}, 2)$ to $\mathcal{C}(k+1)$.
- ▶ It preserves $N(C)$ and $p(C)$.

Therefore, it provides a combinatorial proof of the sum formula.

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- [1] D. M Bradley, *Multiple q -zeta values*, Journal of Algebra, **283** (2005), 752–798.
- [2] B. Brindle, *A unified approach to q MZVs*, Integers, **24** (2024), no. A6, 41.
- [3] T. Suda, 多重ゼータ値の q 類似とその組合せ論的解釈, 東北大学大学院理学研究科修士論文 (2026).
- [4] ———, *Combinatorial proof of the sum formula for q MZVs*, in preparation.